END SEM REVIEW REPORT

Prototyping and Testing Course (DS3001)

Identification Attachment Device for Textile Differentiation

Mentor

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Abstract

The identification of textile materials is a critical requirement in domains such as recycling, quality control, and consumer usage, particularly when garment labels are missing, damaged, or unreliable. This project presents a portable textile identification system that combines mechanical sensing and computer vision into a unified framework. The proposed device operates as a smartphone attachment and integrates triboelectric sensing with macro-level imaging to capture both material-specific and structural characteristics of fabrics.

A mechanically driven rubbing mechanism based on a Scotch yoke is used to generate controlled and repeatable friction between the fabric and a reference surface, producing measurable electrostatic signals. These signals are conditioned and digitized using appropriate electronic circuitry. Simultaneously, a macro lens attached to the smartphone captures high-resolution images of the fabric surface, enabling extraction of texture and weave patterns.

A dataset consisting of macro images and corresponding triboelectric signals is used to train a machine learning model for fabric classification. The system employs a multimodal approach, where both electrical and visual features are combined to improve classification accuracy. The developed prototype demonstrates that this combined sensing strategy provides a reliable, low-cost, and portable alternative to traditional industrial textile identification methods.

1 Introduction

1.1 Problem Statement

Accurate identification of textile materials remains a significant challenge in both consumer and industrial environments. In many real-world scenarios, clothing labels are either missing, faded, or intentionally removed, making it difficult to determine the fabric composition. This creates practical difficulties in garment care, durability assessment, resale evaluation, and sustainable disposal.

The problem is further amplified in textile recycling processes, where large volumes of mixed fabrics need to be sorted based on material type. Since different fabrics require distinct recycling techniques, incorrect identification can reduce efficiency and increase processing costs. Existing solutions typically fall into two categories: manual sorting, which is labor-intensive and prone to error, and advanced industrial systems such as Near-Infrared (NIR) spectroscopy, which are highly accurate but expensive and non-portable.

Therefore, there is a clear need for a system that is both cost-effective and portable while maintaining acceptable levels of accuracy. This project aims to address this gap by developing a compact and intelligent textile identification device.

1.2 Objective

The primary objective of this project is to design and develop a portable textile identification system that can operate without reliance on physical labels. The system aims to:

- Develop a compact and portable device that can be used as a smartphone attachment
- Integrate triboelectric sensing and macro imaging for multi-modal data acquisition
- Capture both electrical and structural characteristics of fabrics
- Implement a machine learning model for automated fabric classification
- Provide real-time identification results through a user-friendly interface

1.3 Assumptions

The design and implementation of the system are based on the following assumptions:

- The fabric sample presented to the system represents the dominant material of the garment
- The surface of the fabric is sufficiently exposed and not heavily contaminated or coated
- The macro imaging system is able to capture clear texture and weave patterns
- Different fabrics generate distinguishable triboelectric responses under controlled friction
- Environmental conditions such as temperature and humidity remain within moderate ranges during operation
- The dataset used for training is representative of the fabric classes considered (cotton, polyester, wool, nylon)

2 Concept Design

The system is based on combining two sensing mechanisms:

- Macro imaging for structural features
- Triboelectric sensing for material properties

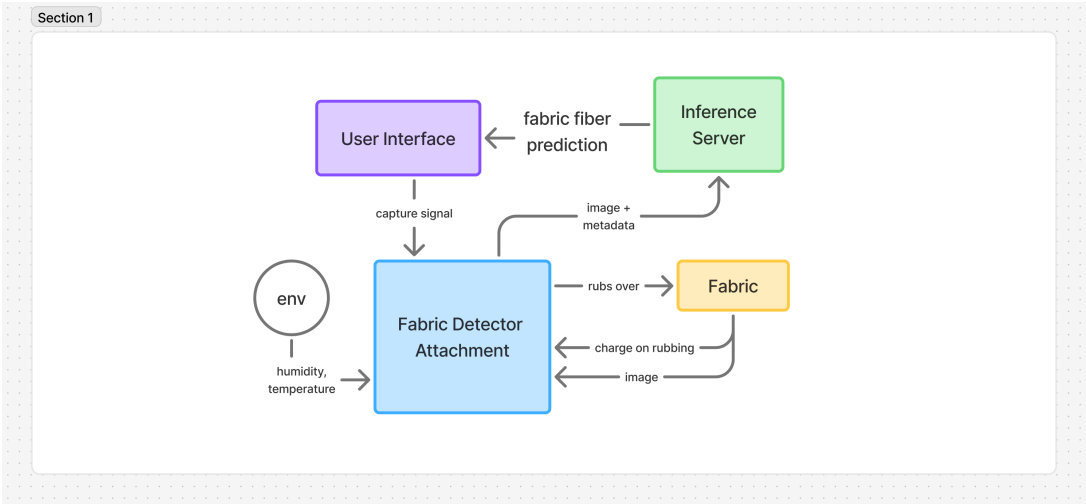

Figure 2.1: Concept Design

3 System Architecture

The system consists of mechanical, electronic, and software subsystems.

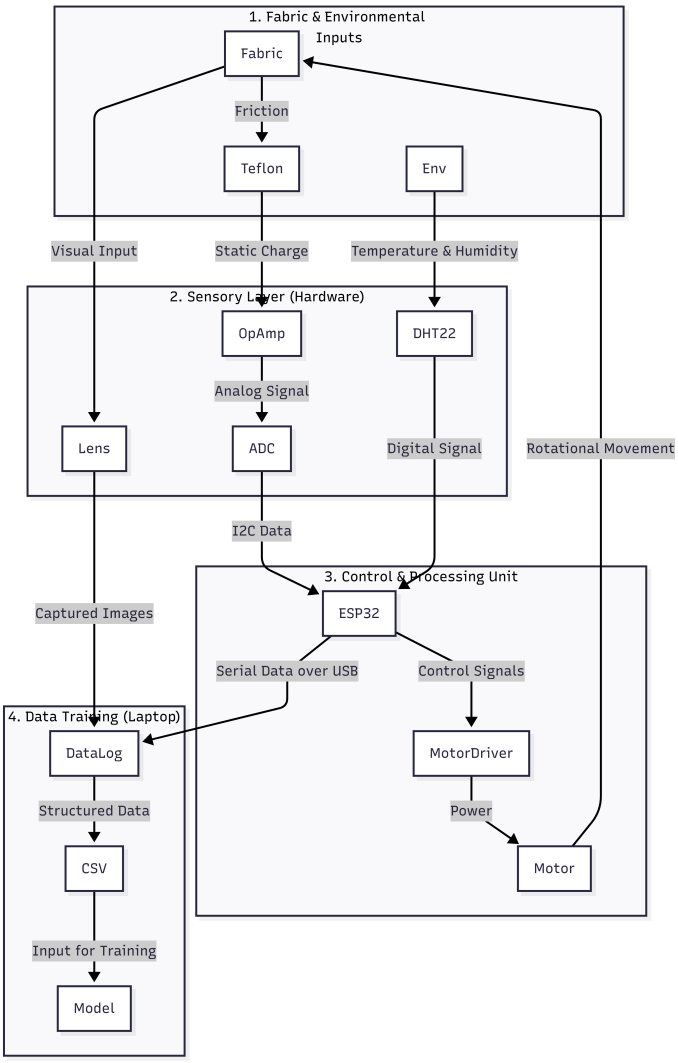

Figure 3.1: Training Phase System Architecture

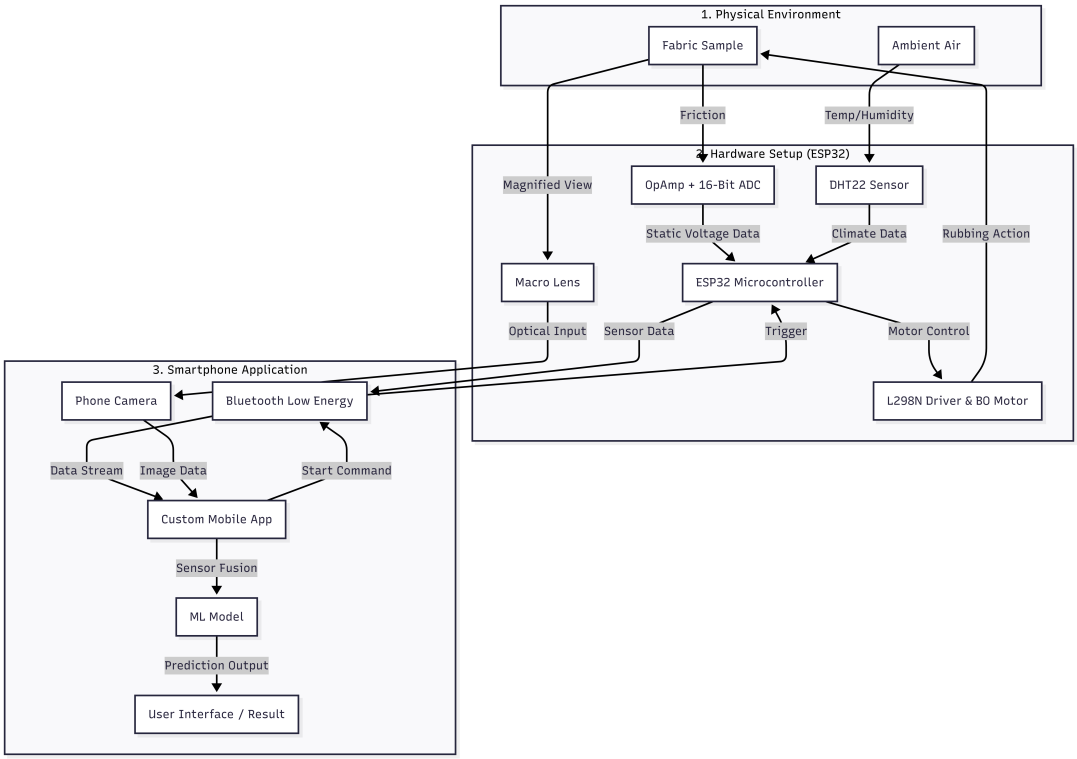

Figure 3.2: Testing Phase System Architecture

3.1 Data Flow

1. Fabric interaction
2. Signal generation
3. Data acquisition
4. Processing
5. Classification

4 Mechanical Design

4.1 Scotch Yoke Mechanism

Used to generate controlled reciprocating motion.

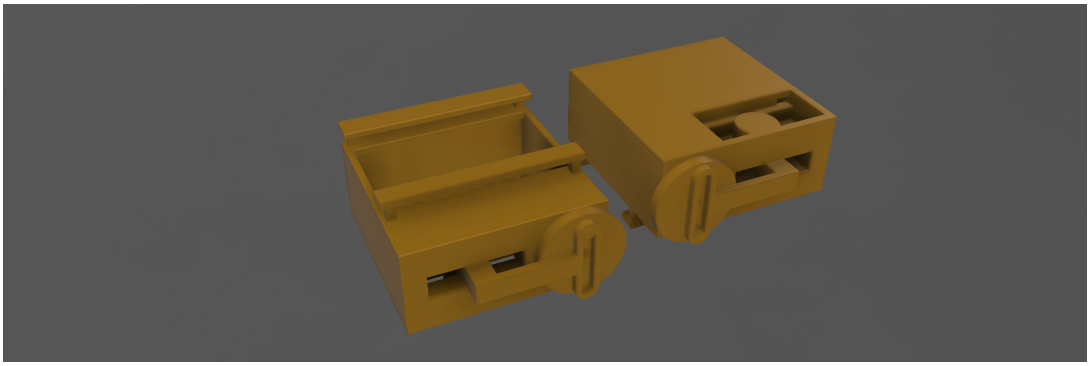

Figure 4.1: Scotch Yoke Mechanism

4.2 Working

Motor rotation is converted into linear motion, enabling consistent friction.

5 Hardware Design

5.1 Components

- ESP32
- TL072 Op-Amp
- ADS1115 ADC
- L298N motor driver
- DC motor
- DHT22 sensor
- Battery + Buck converter

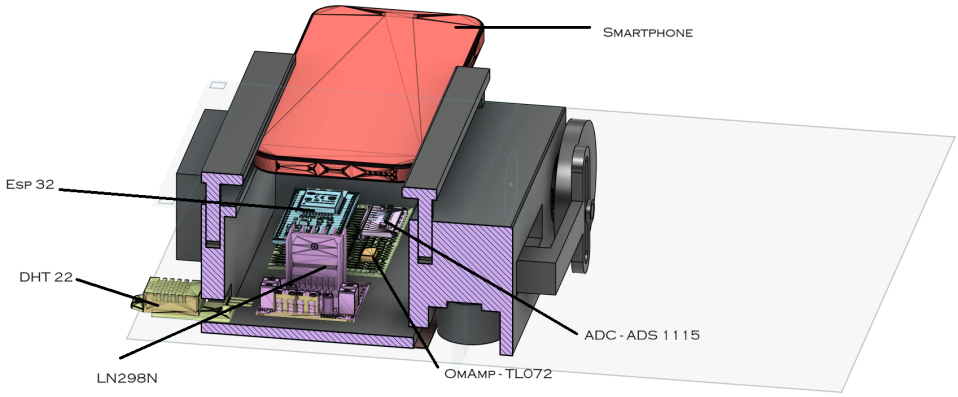

Figure 5.1: Hardware Prototype

6 Software and Machine Learning

6.1 Dataset

- 1000 macro images
- Corresponding voltage signals

6.2 Interface

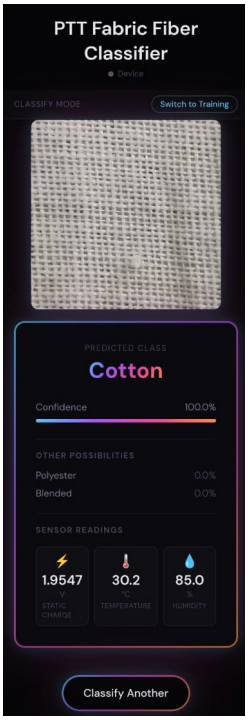

Figure 6.1: Mobile Application Interface

7 Working Methodology

1. Fabric placed in device
2. Motor generates rubbing
3. Charge generated and captured
4. Image captured
5. Data processed
6. Output displayed

8 Prototype Development

Figure 8.1: Prototype Developed

9 Advantages

- Low cost
- Portable
- Real-time results
- Improved accuracy

10 Future Scope

- Development of a more compact design suitable for consumer use
- Integration into a smartphone case form factor
- Improved hardware miniaturization
- Larger dataset and better ML models
- Enhanced user interface and mobile integration
- Potential integration into industrial automation systems

11 Conclusion

This project presents the design and development of a portable textile identification system based on a multi-modal sensing approach. By combining triboelectric signal acquisition with macro-level optical imaging, the system is able to capture both material-specific electrical characteristics and structural features of fabrics.

The implementation of a controlled friction mechanism ensures repeatable signal generation, while the integration of a machine learning model enables automated classification of common fabric types such as cotton, polyester, wool, and nylon. Experimental results demonstrate that the combination of electrical and visual data significantly improves classification reliability compared to using a single sensing modality.

The developed prototype validates the feasibility of a low-cost, compact, and scalable solution for textile identification. The system shows strong potential for applications in textile recycling, quality control, and consumer-level usage.

12 References

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3. Ohi, A. Q., Mridha, M. F., Hamid, M. A., Monowar, M. M., and Kateb, F. A., "FabricNet: A Fiber Recognition Architecture Using Ensemble ConvNets," *IEEE Access*, 2021.